Department of Chemical Engineering, Chulalongkorn University (teaching)

MEA CO2 Capture Ver 1 · operator training simulator

EQUIPMENT LIST AND MECHANICAL DATASHEETS

MEA CO2 capture unit: absorber T-101 / stripper T-102 (30 wt% MEA)

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Originator	Apinan Soottitantawat
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Contents	Cover, sections 1–7: EQL (landscape), 8 datasheets, summaries, GA

Revision record

REV	DATE	DESCRIPTION	ORIGINATOR	CHECKER	APPROVER
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Numbers come from the MEA CO2 Capture Ver 1 simulator model (design point, noise off); items marked TBC/assumed are not design data.

Teaching document for the MEA CO2 capture operator training simulator. Not for construction. Checker and approver are blank: this issue has not been checked or approved.

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1 · Design rules and basis

1. Design pressure (gauge) = the largest of: maximum operating pressure + 10 %; maximum operating pressure + 170 kPa (≈ 25 psi); the relief-valve set pressure that protects the item; the shut-off pressure of the pump or blower that feeds it (maximum suction + 1.2 × rated ΔP). Rounded up to 10 kPa. Rotating machines: casing rated for the shut-off pressure only.

2. Full vacuum (FV) for the T-102 system (T-102, E-102 shell, E-103 shell, V-101) and the E-102 steam side: steam-out and condensing vapour can pull a vacuum (typical for amine regenerators).

3. Design temperature = the larger of maximum operating temperature + 25 K and the trip / steam-out basis (TSHH-206 130 °C, LP-steam steam-out 138.9 °C), rounded up to 5 °C.

4. Shell thickness ASME VIII Div 1 UG-27(c)(1): $t = P \cdot R/(S \cdot E - 0.6 \cdot P) + CA$; 2:1 ellipsoidal head UG-32(d): $t = P \cdot D/(2 \cdot S \cdot E - 0.2 \cdot P) + CA$; R, D inside, corroded; P = design pressure + liquid static head at the bottom; E = 0.85 (spot radiography, assumed).

5. Allowable stress S at the design temperature from typical ASME II-D Table 1A values for SA-240 316L (lower-stress row) and SA-516 Gr 70: verify against the edition in use. Minimum thickness: UG-16(b) 1.5 mm + CA, and a typical handling/rigidity minimum by diameter.

6. Corrosion allowance (assumed): 1.0 mm SS 316L amine and wet-CO2 service, 0 mm plates and pump casings, 1.5 mm CS steam and CW channels. External pressure (FV), wind and seismic loads are not checked here (UG-28, UG-22): HOLD.

7. Materials: SS 316L for every part wetted by amine (lean, rich or loaded) or by wet CO2 (carbonic acid), as the Theory tab states (loaded MEA corrodes carbon steel); carbon steel only where justified: LP-steam / condensate channel of E-102, CW channels, the plate-exchanger frame, the blower base frame.

8. Exchanger area = governing-case required area × 1.15 (typical over-surface) with an assumed service U (incl. fouling): E-101 3.0, E-102 1.1, E-103 0.6, E-104 0.8, E-105 1.0 kW/(m²·K); F for one shell / two tube passes; tubes Ø19.05 mm × 6.1 m, triangular pitch 1.25 do.

Settings used: PSV-301 / PSV-302 set 350 kPa a (assumed, HOLD on CC-PID-101 note 5); PSHH-301 250 kPa a; PIC-301 SP 175, HI alarm 190 kPa a; TSHH-206 130 °C; LP steam 350 kPa a / 138.9 °C and CW 400 kPa a / 30 °C (assumed); K-101 shut-off 118.5 kPa a (model curve). Pump shut-off pressures from MEA-PE-HMB-001 section 6.

Numbers come from the MEA CO2 Capture Ver 1 simulator model (design point, noise off); items marked TBC/assumed are not design data.

2 · Equipment list

Every item of CC-PID-101 rev A and MEA-PFD-001 rev 0; the P&ID draws no filters, reclaimer or sumps (EQ-M07). One row per pressure side; operating = HMB case A.

Tag	Item / type	Size	Side · fluid	Op P kPa a / T °C	Max op P kPa a (basis)	Relief / shut-off basis	Design P kPa g (governs)	Design T °C	FV	Material · CA mm
T-101	Absorber with water wash Column	D 5.4 m, 2 beds × 8.5 m Mellapak Plus 252Y + 3 m wash bed; skirt to 4 m	**Shell** · Flue gas / lean–rich MEA / wash water	107 / 57.3	118 (K-101 shut-off at 100 % speed (118.5 kPa a, model curve))	—	**190** (op + 170 kPa)	**90** (op + 25 K)		SA-240 316L · 1
T-102	Stripper Column	D 2.6 m, 8 m IMTP 50 (≈ 5 stages); skirt to 9.5 m	**Shell** · Rich / lean MEA, CO2 + steam	182 / 120.0	197 (PIC-301 HI 190 kPa a + column ΔP)	PSV-301 set 350 kPa a (assumed, HOLD)	**270** (op + 170 kPa)	**150** (op + 25 K)	FV	SA-240 316L · 1
V-101	Reflux drum Drum (horizontal)	D 1.6 m × 4.0 m T/T, 8 m³ (model K.Vdrum); bottom at 4 m	**Shell** · CO2 + water (carbonic acid)	175 / 40.0	197 (PIC-301 HI 190 kPa a + column ΔP); T 106 °C: CW failure: overhead vapour reaches the drum	PSV-302 set 350 kPa a (assumed; or covered by PSV-301)	**270** (op + 170 kPa)	**140** (steam-out at LP steam (138.9 °C, typical for amine vessels))	FV	SA-240 316L · 1
E-101	Lean/rich exchanger Plate (gasketed or welded, TBC)	530 plates × 2.0 m² (typical large plate, assumed) A 1060 m²	**Rich side** · Rich MEA (flashes at the outlet)	504 / 109.0	504 (P-101 discharge)	P-101 shut-off (677 kPa a)	**580** (P-101 shut-off (677 kPa a))	**150** (op + 25 K)		SA-240 316L · 0
			Lean side · Lean MEA	495 / 120.0	495 (P-102 discharge)	P-102 shut-off (868 kPa a)	**770** (P-102 shut-off (868 kPa a))	**150** (op + 25 K)		SA-240 316L · 0
E-102	Kettle reboiler Kettle (BKU), U-tubes	1582 U-tubes Ø19.05 × 6.1 m straight length (both legs counted in the area); bundle ≈ 1.38 m; shell 3.4 m (model K.Dr) A 1155 m²	**Shell (process)** · Boiling lean MEA / CO2 + steam	182 / 120.0	197 (PIC-301 HI 190 kPa a + column ΔP)	PSV-301 set 350 kPa a (assumed, HOLD)	**270** (op + 170 kPa)	**150** (op + 25 K)	FV	SA-240 316L · 1
			Tubes (steam) · LP steam / condensate	350 / 138.9	350 (LP steam 350 kPa a (assumed header max, TBC))	—	**420** (op + 170 kPa)	**165** (op + 25 K)	FV	SA-240 316L tubes; SA-516 Gr 70 channel · 1.5
E-103	Overhead condenser Shell and tube, 1 shell / 2 tube passes (BEM)	877 tubes Ø19.05 × 6.1 m, 2 passes; bundle ≈ 0.77 m; shell ID ≈ 0.85 m A 320 m²	**Shell (process)** · CO2 + steam, condensing	181 / 103.1	197 (PIC-301 HI 190 kPa a + column ΔP)	PSV-301 set 350 kPa a (assumed, HOLD)	**270** (op + 170 kPa)	**140** (steam-out at LP steam (138.9 °C, typical for amine vessels))	FV	SA-240 316L · 1
			Tubes (CW) · Cooling water	400 / 30.0	400 (CW supply 400 kPa a (assumed, TBC))	—	**470** (op + 170 kPa)	**70** (op + 25 K)		SA-240 316L tubes; CS channel · 1.5
E-104	Lean amine cooler Shell and tube, 1 shell / 2 tube passes (BEM)	1480 tubes Ø19.05 × 6.1 m, 2 passes; bundle ≈ 0.98 m; shell ID ≈ 1.05 m A 540 m²	**Shell (lean MEA)** · Lean MEA	453 / 51.6	495 (P-102 discharge)	P-102 shut-off (868 kPa a)	**770** (P-102 shut-off (868 kPa a))	**85** (op + 25 K)		SA-240 316L · 1
			Tubes (CW) · Cooling water	400 / 30.0	400 (CW supply 400 kPa a (assumed, TBC))	—	**470** (op + 170 kPa)	**70** (op + 25 K)		SA-240 316L tubes; CS channel · 1.5
E-105	Wash-water cooler Shell and tube, 1 shell / 2 tube passes (BEM)	2877 tubes Ø19.05 × 6.1 m, 2 passes; bundle ≈ 1.32 m; shell ID ≈ 1.40 m A 1050 m²	**Shell (wash water)** · Wash water (traces of MEA)	468 / 47.9	468 (P-104 discharge)	P-104 shut-off (512 kPa a)	**540** (op + 170 kPa)	**80** (op + 25 K)		SA-240 316L · 1
			Tubes (CW) · Cooling water	400 / 30.0	400 (CW supply 400 kPa a (assumed, TBC))	—	**470** (op + 170 kPa)	**70** (op + 25 K)		SA-240 316L tubes; CS channel · 1.5
P-101	Rich amine pump Centrifugal pump, OH2 (API 610 type, TBC)	487 m³/h rated × 37.9 m; motor 90 kW; NPSHa 12.2 m	**Casing** · Rich MEA	504 / 43.6	504	max suction + shut-off ΔP (677 kPa a)	**580** (max suction + shut-off ΔP (677 kPa a))	**70** (op + 25 K)		CF3M (cast 316L) · 0
P-102	Lean amine pump Centrifugal pump, OH2 (API 610 type, TBC)	510 m³/h rated × 36.9 m; motor 90 kW; NPSHa 5.2 m	**Casing** · Lean MEA at its bubble point	495 / 120.0	495	max suction + shut-off ΔP (868 kPa a)	**770** (max suction + shut-off ΔP (868 kPa a))	**145** (op + 25 K)		CF3M (cast 316L) · 0
P-103	Reflux pump Centrifugal pump, OH2 (API 610 type, TBC)	7 m³/h rated × 27.2 m; motor 2.2 kW; NPSHa 3.7 m	**Casing** · Water saturated with CO2	473 / 40.0	473	max suction + shut-off ΔP (713 kPa a)	**620** (max suction + shut-off ΔP (713 kPa a))	**70** (op + 25 K)		CF3M (cast 316L) · 0
P-104	Wash water pump Centrifugal pump, OH2 (API 610 type, TBC)	508 m³/h rated × 9.3 m; motor 22 kW; NPSHa 38.5 m	**Casing** · Wash water (traces of MEA)	468 / 47.9	468	max suction + shut-off ΔP (512 kPa a)	**420** (max suction + shut-off ΔP (512 kPa a))	**75** (op + 25 K)		CF3M (cast 316L) · 0
K-101	Flue gas blower Centrifugal blower, VFD	201 × 10³ m³/h at 4.7 kPa; 313 kW shaft at 80 % speed (model)	**Casing** · Wet flue gas (saturated, 40 °C)	107 / 40.0	118	shut-off at 100 % speed (118.5 kPa a)	**20** (shut-off at 100 % speed (118.5 kPa a))	**70** (op + 25 K)		SS 304L (TBC) · 0

3 · Datasheets

Datasheet T-101 — Absorber with water wash

MEA-ME-EQL-001-T-101 rev 0	**Column**
Service / drawings	amine · CC-PID-101 rev A · MEA-PFD-001 rev 0 · HMB MEA-PE-HMB-001 rev 0
Size	D 5.4 m, 2 beds × 8.5 m Mellapak Plus 252Y + 3 m wash bed; skirt to 4 m
Design data	**Shell**
Fluid	Flue gas / lean–rich MEA / wash water
Operating P kPa a / T °C (case A)	107 / 57.3
Max operating P kPa a / T °C	118 / 64.1
Max operating basis	K-101 shut-off at 100 % speed (118.5 kPa a, model curve)
Relief / shut-off basis	—
Design pressure, kPa g	**190** (op + 170 kPa)
Design temperature, °C	**90** (op + 25 K)
Material / CA mm	SA-240 316L / 1
Code	ASME VIII Div 1
Wall thickness	
Pressure at the bottom / S	217.3 kPa g (incl. liquid head) · S = 98.2 MPa at 90 °C (SA-240 316L, typical) · E 0.85
Shell t = PR/(SE − 0.6P) + CA	8.04 mm → nominal **14 mm** (rigidity (typical minimum); rigidity minimum 14 mm)
2:1 head t = PD/(2SE − 0.2P) + CA	8.03 mm → nominal **10 mm**
Hydrotest (UG-99)	290 kPa g
Materials and notes	
Materials	SS 316L shell, internals and packing
Overpressure protection	none on CC-PID-101 (open to the stack); K-101 shut-off sets the design pressure
Holds	External pressure (FV), wind, seismic and nozzle loads: HOLD. Items marked TBC/assumed are not design data.

Datasheet T-102 — Stripper

MEA-ME-EQL-001-T-102 rev 0	**Column**
Service / drawings	amine · CC-PID-101 rev A · MEA-PFD-001 rev 0 · HMB MEA-PE-HMB-001 rev 0
Size	D 2.6 m, 8 m IMTP 50 (≈ 5 stages); skirt to 9.5 m
Design data	**Shell**
Fluid	Rich / lean MEA, CO2 + steam
Operating P kPa a / T °C (case A)	182 / 120.0
Max operating P kPa a / T °C	197 / 120.0
Max operating basis	PIC-301 HI 190 kPa a + column ΔP
Relief / shut-off basis	PSV-301 set 350 kPa a (assumed, HOLD)
Design pressure, kPa g	**270** / FV (op + 170 kPa)
Design temperature, °C	**150** (op + 25 K)
Material / CA mm	SA-240 316L / 1
Code	ASME VIII Div 1
Wall thickness	
Pressure at the bottom / S	290.4 kPa g (incl. liquid head) · S = 87.5 MPa at 150 °C (SA-240 316L, typical) · E 0.85
Shell t = PR/(SE − 0.6P) + CA	6.09 mm → nominal **10 mm** (rigidity (typical minimum); rigidity minimum 10 mm)
2:1 head t = PD/(2SE − 0.2P) + CA	6.08 mm → nominal **8 mm**
Hydrotest (UG-99)	462 kPa g
Materials and notes	
Materials	SS 316L shell, internals and packing
Overpressure protection	PSV-301 on the overhead (HOLD), PSHH-301 250 kPa a (I-2)
Holds	External pressure (FV), wind, seismic and nozzle loads: HOLD. Items marked TBC/assumed are not design data.

Datasheet V-101 — Reflux drum

MEA-ME-EQL-001-V-101 rev 0	**Drum (horizontal)**
Service / drawings	wet CO2 · CC-PID-101 rev A · MEA-PFD-001 rev 0 · HMB MEA-PE-HMB-001 rev 0
Size	D 1.6 m × 4.0 m T/T, 8 m³ (model K.Vdrum); bottom at 4 m
Design data	**Shell**
Fluid	CO2 + water (carbonic acid)
Operating P kPa a / T °C (case A)	175 / 40.0
Max operating P kPa a / T °C	197 / 106.1
Max operating basis	PIC-301 HI 190 kPa a + column ΔP; CW failure: overhead vapour reaches the drum
Relief / shut-off basis	PSV-302 set 350 kPa a (assumed; or covered by PSV-301)
Design pressure, kPa g	**270** / FV (op + 170 kPa)
Design temperature, °C	**140** (steam-out at LP steam (138.9 °C, typical for amine vessels))
Material / CA mm	SA-240 316L / 1
Code	ASME VIII Div 1
Wall thickness	
Pressure at the bottom / S	285.7 kPa g (incl. liquid head) · S = 89.1 MPa at 140 °C (SA-240 316L, typical) · E 0.85
Shell t = PR/(SE − 0.6P) + CA	4.03 mm → nominal **7 mm** (rigidity (typical minimum); rigidity minimum 7 mm)
2:1 head t = PD/(2SE − 0.2P) + CA	4.02 mm → nominal **5 mm**
Hydrotest (UG-99)	453 kPa g
Materials and notes	
Materials	SS 316L
Overpressure protection	PSV-302 (HOLD) or PSV-301 if V-101 cannot be isolated from T-102
Holds	External pressure (FV), wind, seismic and nozzle loads: HOLD. Items marked TBC/assumed are not design data.

Datasheet E-101 — Lean/rich exchanger

MEA-ME-EQL-001-E-101 rev 0	**Plate (gasketed or welded, TBC)**	
Service / drawings	amine · CC-PID-101 rev A · MEA-PFD-001 rev 0 · HMB MEA-PE-HMB-001 rev 0	
Size	530 plates × 2.0 m² (typical large plate, assumed)	
Design data	**Rich side**	**Lean side**
Fluid	Rich MEA (flashes at the outlet)	Lean MEA
Operating P kPa a / T °C (case A)	504 / 109.0	495 / 120.0
Max operating P kPa a / T °C	504 / 120.0	495 / 120.0
Max operating basis	P-101 discharge	P-102 discharge
Relief / shut-off basis	P-101 shut-off (677 kPa a)	P-102 shut-off (868 kPa a)
Design pressure, kPa g	**580** (P-101 shut-off (677 kPa a))	**770** (P-102 shut-off (868 kPa a))
Design temperature, °C	**150** (op + 25 K)	**150** (op + 25 K)
Material / CA mm	SA-240 316L / 0	SA-240 316L / 0
Code	ASME VIII Div 1 (plate exchanger: vendor code, TBC)	ASME VIII Div 1 (plate exchanger: vendor code, TBC)
Thermal rating		
Duty case A / governing	18.45 MW / 30.09 MW (case B)	
Hot side (case A)	lean MEA 120.0 → 51.6 °C	
Cold side (case A)	rich MEA 43.6 → 109.0 °C	
LMTD · F · U (assumed)	9.43 K · 1.000 · 3 kW/(m²·K)	
Area required A / datasheet	652 m² / **1060 m²** (case B × 1.15)	
Rating U·A·F·LMTD vs duty	A: 29.98 vs 18.45 MW (63 %) · B: 34.91 vs 30.09 MW (16 %) · C: 25.84 vs 12.19 MW (112 %)	
Geometry	530 plates × 2.0 m² (typical large plate, assumed)	
Simulator UA vs datasheet U·A	2000 vs 3180 kW/K (63 %)	
Materials and notes		
Materials	SS 316L plates; carbon-steel frame (not wetted)	
Overpressure protection	none; both sides pump-fed — design for pump shut-off	
Holds	External pressure (FV), wind, seismic and nozzle loads: HOLD. Items marked TBC/assumed are not design data.	

Datasheet E-102 — Kettle reboiler

MEA-ME-EQL-001-E-102 rev 0	**Kettle (BKU), U-tubes**	
Service / drawings	amine · CC-PID-101 rev A · MEA-PFD-001 rev 0 · HMB MEA-PE-HMB-001 rev 0	
Size	1582 U-tubes Ø19.05 × 6.1 m straight length (both legs counted in the area); bundle ≈ 1.38 m; shell 3.4 m (model K.Dr)	
Design data	**Shell (process)**	**Tubes (steam)**
Fluid	Boiling lean MEA / CO2 + steam	LP steam / condensate
Operating P kPa a / T °C (case A)	182 / 120.0	350 / 138.9
Max operating P kPa a / T °C	197 / 120.0	350 / 138.9
Max operating basis	PIC-301 HI 190 kPa a + column ΔP	LP steam 350 kPa a (assumed header max, TBC)
Relief / shut-off basis	PSV-301 set 350 kPa a (assumed, HOLD)	—
Design pressure, kPa g	**270** / FV (op + 170 kPa)	**420** / FV (op + 170 kPa)
Design temperature, °C	**150** (op + 25 K)	**165** (op + 25 K)
Material / CA mm	SA-240 316L / 1	SA-240 316L tubes; SA-516 Gr 70 channel / 1.5
Code	ASME VIII Div 1	ASME VIII Div 1
Wall thickness		
Pressure at the bottom / S	270.0 kPa g (incl. liquid head) · S = 87.5 MPa at 150 °C (SA-240 316L, typical) · E 0.85	
Shell t = PR/(SE − 0.6P) + CA	7.19 mm → nominal **12 mm** (rigidity (typical minimum); rigidity minimum 12 mm)	
2:1 head t = PD/(2SE − 0.2P) + CA	7.18 mm → nominal **8 mm**	
Hydrotest (UG-99)	462 kPa g	
Thermal rating		
Duty case A / governing	13.69 MW / 20.86 MW (case B)	
Hot side (case A)	LP steam condensing 138.9 → 138.9 °C	
Cold side (case A)	lean MEA boiling 120.0 → 120.0 °C	
LMTD · F · U (assumed)	18.90 K · 1.000 · 1.1 kW/(m²·K)	
Area required A / datasheet	658 m² / **1155 m²** (case B × 1.15)	
Rating U·A·F·LMTD vs duty	A: 24.01 vs 13.69 MW (75 %) · B: 24.00 vs 20.86 MW (15 %) · C: 24.01 vs 10.12 MW (137 %)	
Geometry	1582 U-tubes Ø19.05 × 6.1 m straight length (both legs counted in the area); bundle ≈ 1.38 m; shell 3.4 m (model K.Dr)	
Simulator UA vs datasheet U·A	1400 vs 1270 kW/K (110 %)	
Tube rupture (10/13 rule)	Tubes (steam) 420 vs Shell (process) 270 kPa g → **rupture is a relief case**	
Materials and notes		
Materials	SS 316L shell and tubes (amine side); CS steam channel (steam/condensate only)	
Overpressure protection	process side: PSV-301 via T-102 (HOLD); steam side PSV-207 (HOLD, P&ID note 5)	
Holds	External pressure (FV), wind, seismic and nozzle loads: HOLD. Items marked TBC/assumed are not design data.	

Datasheet E-103 — Overhead condenser

MEA-ME-EQL-001-E-103 rev 0	**Shell and tube, 1 shell / 2 tube passes (BEM)**	
Service / drawings	wet CO2 · CC-PID-101 rev A · MEA-PFD-001 rev 0 · HMB MEA-PE-HMB-001 rev 0	
Size	877 tubes Ø19.05 × 6.1 m, 2 passes; bundle ≈ 0.77 m; shell ID ≈ 0.85 m	
Design data	**Shell (process)**	**Tubes (CW)**
Fluid	CO2 + steam, condensing	Cooling water
Operating P kPa a / T °C (case A)	181 / 103.1	400 / 30.0
Max operating P kPa a / T °C	197 / 106.1	400 / 45.0
Max operating basis	PIC-301 HI 190 kPa a + column ΔP	CW supply 400 kPa a (assumed, TBC)
Relief / shut-off basis	PSV-301 set 350 kPa a (assumed, HOLD)	—
Design pressure, kPa g	**270** / FV (op + 170 kPa)	**470** (op + 170 kPa)
Design temperature, °C	**140** (steam-out at LP steam (138.9 °C, typical for amine vessels))	**70** (op + 25 K)
Material / CA mm	SA-240 316L / 1	SA-240 316L tubes; CS channel / 1.5
Code	ASME VIII Div 1	ASME VIII Div 1
Wall thickness		
Pressure at the bottom / S	270.0 kPa g (incl. liquid head) · S = 89.1 MPa at 140 °C (SA-240 316L, typical) · E 0.85	
Shell t = PR/(SE − 0.6P) + CA	2.52 mm → nominal **5 mm** (rigidity (typical minimum); rigidity minimum 5 mm)	
2:1 head t = PD/(2SE − 0.2P) + CA	2.52 mm → nominal **3 mm**	
Hydrotest (UG-99)	453 kPa g	
Thermal rating		
Duty case A / governing	4.08 MW / 5.03 MW (case B)	
Hot side (case A)	overhead vapour 103.1 → 40.0 °C	
Cold side (case A)	cooling water 30.0 → 34.4 °C	
LMTD · F · U (assumed)	30.46 K · 0.936 · 0.6 kW/(m²·K)	
Area required A / datasheet	239 m² / **320 m²** (case B × 1.15)	
Rating U·A·F·LMTD vs duty	A: 5.47 vs 4.08 MW (34 %) · B: 5.81 vs 5.03 MW (16 %) · C: 5.44 vs 3.76 MW (45 %)	
Geometry	877 tubes Ø19.05 × 6.1 m, 2 passes; bundle ≈ 0.77 m; shell ID ≈ 0.85 m	
Simulator UA vs datasheet U·A	130 vs 192 kW/K (68 %)	
Tube rupture (10/13 rule)	Tubes (CW) 470 vs Shell (process) 270 kPa g → **rupture is a relief case**	
Materials and notes		
Materials	SS 316L shell and tubes (wet CO2); CS CW channel	
Overpressure protection	process side via PSV-301 (HOLD)	
Holds	External pressure (FV), wind, seismic and nozzle loads: HOLD. Items marked TBC/assumed are not design data.	

Datasheet E-104 — Lean amine cooler

MEA-ME-EQL-001-E-104 rev 0	**Shell and tube, 1 shell / 2 tube passes (BEM)**	
Service / drawings	amine · CC-PID-101 rev A · MEA-PFD-001 rev 0 · HMB MEA-PE-HMB-001 rev 0	
Size	1480 tubes Ø19.05 × 6.1 m, 2 passes; bundle ≈ 0.98 m; shell ID ≈ 1.05 m	
Design data	**Shell (lean MEA)**	**Tubes (CW)**
Fluid	Lean MEA	Cooling water
Operating P kPa a / T °C (case A)	453 / 51.6	400 / 30.0
Max operating P kPa a / T °C	495 / 56.4	400 / 45.0
Max operating basis	P-102 discharge	CW supply 400 kPa a (assumed, TBC)
Relief / shut-off basis	P-102 shut-off (868 kPa a)	—
Design pressure, kPa g	**770** (P-102 shut-off (868 kPa a))	**470** (op + 170 kPa)
Design temperature, °C	**85** (op + 25 K)	**70** (op + 25 K)
Material / CA mm	SA-240 316L / 1	SA-240 316L tubes; CS channel / 1.5
Code	ASME VIII Div 1	ASME VIII Div 1
Wall thickness		
Pressure at the bottom / S	770.0 kPa g (incl. liquid head) · S = 99.8 MPa at 85 °C (SA-240 316L, typical) · E 0.85	
Shell t = PR/(SE − 0.6P) + CA	5.80 mm → nominal **7 mm** (rigidity (typical minimum); rigidity minimum 7 mm)	
2:1 head t = PD/(2SE − 0.2P) + CA	5.78 mm → nominal **6 mm**	
Hydrotest (UG-99)	1154 kPa g	
Thermal rating		
Duty case A / governing	3.13 MW / 6.65 MW (case B)	
Hot side (case A)	lean MEA 51.6 → 40.0 °C	
Cold side (case A)	cooling water 30.0 → 31.5 °C	
LMTD · F · U (assumed)	14.48 K · 0.986 · 0.8 kW/(m²·K)	
Area required A / datasheet	274 m² / **540 m²** (case B × 1.15)	
Rating U·A·F·LMTD vs duty	A: 6.17 vs 3.13 MW (97 %) · B: 7.64 vs 6.65 MW (15 %) · C: 5.69 vs 1.66 MW (242 %)	
Geometry	1480 tubes Ø19.05 × 6.1 m, 2 passes; bundle ≈ 0.98 m; shell ID ≈ 1.05 m	
Simulator UA vs datasheet U·A	220 vs 432 kW/K (51 %)	
Tube rupture (10/13 rule)	Shell (lean MEA) 770 vs Tubes (CW) 470 kPa g → **rupture is a relief case**	
Materials and notes		
Materials	SS 316L shell and tubes (amine); CS CW channel	
Overpressure protection	none on CC-PID-101: thermal relief on the CW side if it can be blocked in (TBC)	
Holds	External pressure (FV), wind, seismic and nozzle loads: HOLD. Items marked TBC/assumed are not design data.	

Datasheet E-105 — Wash-water cooler

MEA-ME-EQL-001-E-105 rev 0	**Shell and tube, 1 shell / 2 tube passes (BEM)**	
Service / drawings	amine traces · CC-PID-101 rev A · MEA-PFD-001 rev 0 · HMB MEA-PE-HMB-001 rev 0	
Size	2877 tubes Ø19.05 × 6.1 m, 2 passes; bundle ≈ 1.32 m; shell ID ≈ 1.40 m	
Design data	**Shell (wash water)**	**Tubes (CW)**
Fluid	Wash water (traces of MEA)	Cooling water
Operating P kPa a / T °C (case A)	468 / 47.9	400 / 30.0
Max operating P kPa a / T °C	468 / 52.6	400 / 45.0
Max operating basis	P-104 discharge	CW supply 400 kPa a (assumed, TBC)
Relief / shut-off basis	P-104 shut-off (512 kPa a)	—
Design pressure, kPa g	**540** (op + 170 kPa)	**470** (op + 170 kPa)
Design temperature, °C	**80** (op + 25 K)	**70** (op + 25 K)
Material / CA mm	SA-240 316L / 1	SA-240 316L tubes; CS channel / 1.5
Code	ASME VIII Div 1	ASME VIII Div 1
Wall thickness		
Pressure at the bottom / S	540.0 kPa g (incl. liquid head) · S = 101.4 MPa at 80 °C (SA-240 316L, typical) · E 0.85	
Shell t = PR/(SE − 0.6P) + CA	5.41 mm → nominal **7 mm** (rigidity (typical minimum); rigidity minimum 7 mm)	
2:1 head t = PD/(2SE − 0.2P) + CA	5.39 mm → nominal **6 mm**	
Hydrotest (UG-99)	797 kPa g	
Thermal rating		
Duty case A / governing	5.77 MW / 8.24 MW (case B)	
Hot side (case A)	wash water 47.9 → 37.0 °C	
Cold side (case A)	cooling water 30.0 → 35.5 °C	
LMTD · F · U (assumed)	9.42 K · 0.871 · 1 kW/(m²·K)	
Area required A / datasheet	703 m² / **1050 m²** (case B × 1.15)	
Rating U·A·F·LMTD vs duty	A: 8.62 vs 5.77 MW (49 %) · B: 9.49 vs 8.24 MW (15 %) · C: 8.64 vs 4.28 MW (102 %)	
Geometry	2877 tubes Ø19.05 × 6.1 m, 2 passes; bundle ≈ 1.32 m; shell ID ≈ 1.40 m	
Simulator UA vs datasheet U·A	no UA in the model (sized here)	
Tube rupture (10/13 rule)	Shell (wash water) 540 vs Tubes (CW) 470 kPa g → no rupture case	
Materials and notes		
Materials	SS 316L shell and tubes; CS CW channel	
Overpressure protection	none on CC-PID-101	
Holds	External pressure (FV), wind, seismic and nozzle loads: HOLD. Items marked TBC/assumed are not design data.	

4 · Wall thickness summary (ASME VIII Div 1)

Item	D m	P design kPa g (+ head)	T °C	S MPa	t shell req. mm	t head req. mm	UG-16 min mm	Rigidity min mm	Shell / head nominal mm	Governs	Hydrotest kPa g
T-101	5.40	190 (217)	90	98.2	8.04	8.03	2.5	14	**14 / 10**	rigidity (typical minimum)	290
T-102	2.60	270 (290)	150	87.5	6.09	6.08	2.5	10	**10 / 8**	rigidity (typical minimum)	462
V-101	1.60	270 (286)	140	89.1	4.03	4.02	2.5	7	**7 / 5**	rigidity (typical minimum)	453
E-102	3.40	270 (270)	150	87.5	7.19	7.18	2.5	12	**12 / 8**	rigidity (typical minimum)	462
E-103	0.85	270 (270)	140	89.1	2.52	2.52	2.5	5	**5 / 3**	rigidity (typical minimum)	453
E-104	1.05	770 (770)	85	99.8	5.80	5.78	2.5	7	**7 / 6**	rigidity (typical minimum)	1154
E-105	1.40	540 (540)	80	101.4	5.41	5.39	2.5	7	**7 / 6**	rigidity (typical minimum)	797

t = P·R/(S·E − 0.6·P) + CA (UG-27(c)(1)); heads t = P·D/(2·S·E − 0.2·P) + CA (UG-32(d)); E 0.85. S: typical ASME II-D values, verify. External pressure (FV), wind and seismic not checked: HOLD (EQ-M06).

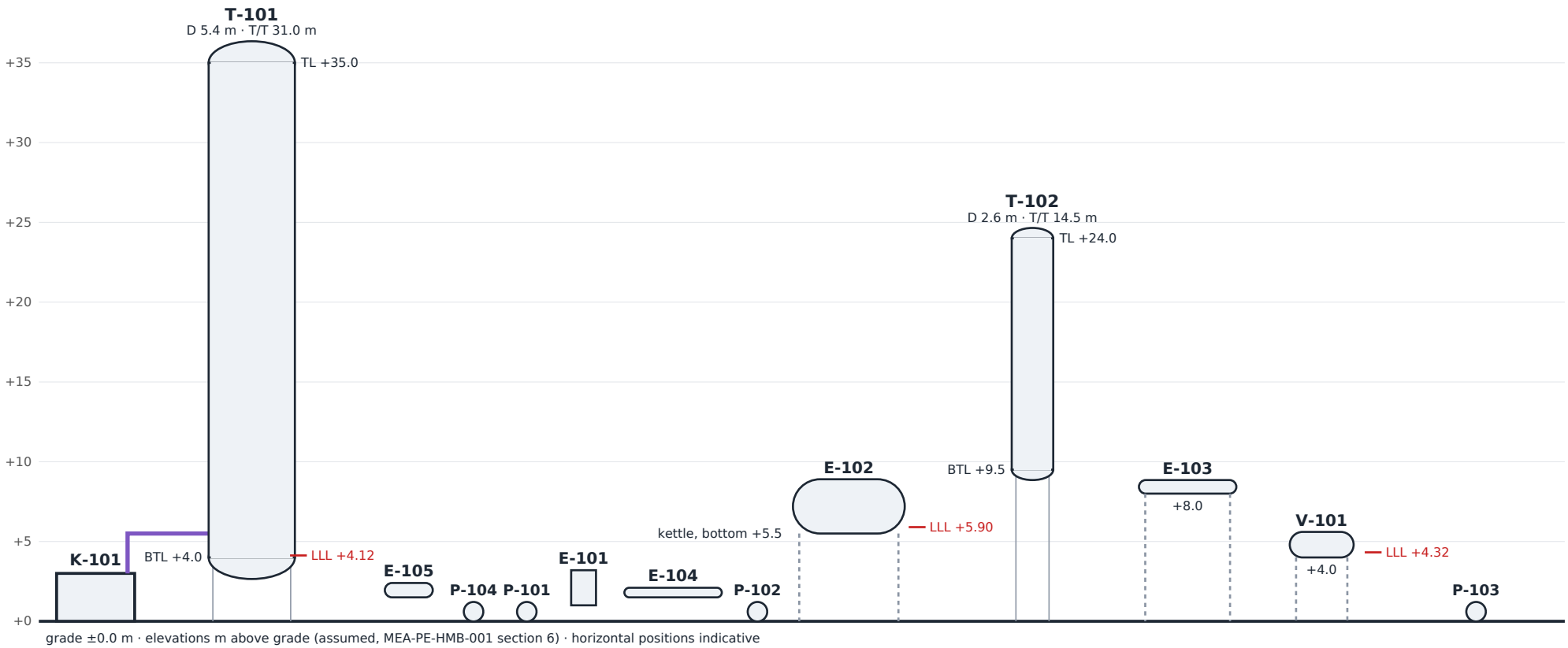
5 · Exchanger rating summary

Exchanger	Case	Duty MW	Hot in → out °C	Cold in → out °C	LMTD K	F	U kW/m²K	A req m²	A datasheet m²	U·A·F·LMTD MW	Margin %
E-101	A	18.45	120.0 → 51.6	43.6 → 109.0	9.43	1.000	3	652	1060	29.98	63
E-101	B (gov.)	30.09	120.0 → 56.4	46.7 → 107.7	10.98	1.000	3	914	1060	34.91	16
E-101	C	12.19	120.0 → 49.6	43.1 → 110.0	8.13	1.000	3	500	1060	25.84	112
E-102	A	13.69	138.9 → 138.9	120.0 → 120.0	18.90	1.000	1.1	658	1155	24.01	75
E-102	B (gov.)	20.86	138.9 → 138.9	120.0 → 120.0	18.89	1.000	1.1	1004	1155	24.00	15
E-102	C	10.12	138.9 → 138.9	120.0 → 120.0	18.90	1.000	1.1	487	1155	24.01	137
E-103	A	4.08	103.1 → 40.0	30.0 → 34.4	30.46	0.936	0.6	239	320	5.47	34
E-103	B (gov.)	5.03	101.1 → 41.1	30.0 → 33.0	31.41	0.963	0.6	277	320	5.81	16
E-103	C	3.76	106.1 → 40.0	30.0 → 35.6	30.95	0.915	0.6	221	320	5.44	45
E-104	A	3.13	51.6 → 40.0	30.0 → 31.5	14.48	0.986	0.8	274	540	6.17	97
E-104	B (gov.)	6.65	56.4 → 42.4	30.0 → 31.6	17.91	0.988	0.8	470	540	7.64	15
E-104	C	1.66	49.6 → 40.0	30.0 → 32.1	13.42	0.981	0.8	158	540	5.69	242
E-105	A	5.77	47.9 → 37.0	30.0 → 35.5	9.42	0.871	1	703	1050	8.62	49
E-105	B (gov.)	8.24	52.6 → 37.0	30.0 → 36.5	10.90	0.829	1	913	1050	9.49	15
E-105	C	4.28	45.1 → 37.0	30.0 → 34.0	8.88	0.927	1	520	1050	8.64	102

Tube-rupture screen (API 521 "10/13 rule")

Exchanger	High-pressure side	Low-pressure side	1.3 × DP low ≥ DP high?	Result
E-102	Tubes (steam) 420 kPa g	Shell (process) 270 kPa g	351 vs 420	**tube rupture is a relief case**
E-103	Tubes (CW) 470 kPa g	Shell (process) 270 kPa g	351 vs 470	**tube rupture is a relief case**
E-104	Shell (lean MEA) 770 kPa g	Tubes (CW) 470 kPa g	611 vs 770	**tube rupture is a relief case**
E-105	Shell (wash water) 540 kPa g	Tubes (CW) 470 kPa g	611 vs 540	no rupture case

6 · GA / elevation sketch



Elevations are those the pump hydraulics need: T-101 skirt for P-101, the kettle E-102 on a structure for the NPSH of P-102 on boiling lean, T-102 above the kettle so its sump drains into it, V-101 above P-103. Plot plan and piping: Layout tab.

7 · Engineering queries

EQ	What	Documents	Finding	Suggested resolution
EQ-M01	E-102 tube rupture is a relief case	MEA-ME-EQL-001 vs relief basis (HOLD)	Steam side 420 kPa g vs process side 270 kPa g: 1.3 × 270 = 351 < 420, so the 10/13 rule (API 521) does not exclude a tube rupture; steam would flow into the T-102 system.	Add tube rupture to PSV-301 sizing (Relief & vent tab) or raise the T-102 system design pressure to ≥ 330 kPa g.
EQ-M02	E-103 and E-104 tube rupture	EQL vs CW system (assumed 400 kPa a)	E-103: CW 470 kPa g vs process 270 kPa g (CW into the T-102 system). E-104: lean side 770 kPa g vs CW 470 kPa g (amine into the CW return).	Relief cases for PSV-301 and for the CW return of E-104, or raise the low-side design pressures; confirm the CW supply pressure.
EQ-M03	PSV-301 set vs blocked-outlet pressure	CC-PID-101 note 5 (HOLD), SRS MEA-IN-SRS-001, EQL	With every trip disabled the model reaches ≥ 400 kPa a on a blocked CO2 outlet (189 s to 350 kPa a); the T-102 system MAWP 270 kPa g allows 398 kPa a at 10 % accumulation. PSHH-301 / I-2 alone does not stop the rise (SRS SIF-2 gap).	Size PSV-301 for blocked outlet with the hot rich feed flashing (Relief & vent tab); keep set ≤ MAWP; add SIF-2R (stop P-101) as the LOPA recommends.
EQ-M04	Materials: P&ID note 6 vs the equipment list	CC-PID-101 note 6 vs MEA-ME-EQL-001, Theory tab	Note 6 allows "CS + CA for cold lean service"; the EQL uses SS 316L for every amine-wetted part (E-104 shell, P-102 casing, T-101 shell) as the Theory tab states.	Keep SS 316L; revise note 6 at rev B, or justify CS + CA by a corrosion study (loading, velocity, heat-stable salts).
EQ-M05	Which side of E-102 the HOLD PSV protects	CC-PID-101 note 5 vs EQL, Safety layers register	Note 5 lists a PSV on the "E-102 shell". In a kettle the shell is the process side, open to T-102 and covered by PSV-301; the steam tubes need their own PSV-207.	Confirm PSV-207 on the steam side; delete the shell PSV or record it as PSV-301.
EQ-M06	Mechanical checks not done	EQL	Full vacuum (external pressure, stiffening rings), wind and seismic on T-101 (35 m) and T-102 (24 m), nozzle loads and the T-101 skirt are not checked; T-101 thickness is set by the rigidity minimum.	HOLD for the vessel vendor; check UG-28 for the FV items before issue.
EQ-M07	Equipment not on the P&ID	CC-PID-101 note 7 vs EQL	No solvent filters (mechanical, carbon), reclaimer, amine storage or drain sump are drawn, so the EQL has none; an MEA unit needs them (foaming and degradation are faults in the simulator).	Add them at P&ID rev B and to the EQL; the filter ΔP then enters the P-101 hydraulics.
EQ-M08	Model UA vs datasheet U·A	src/model.js K vs EQL	E-101 model UA 2000 kW/K = 63 % of the datasheet U·A (sized for the 6 % CO2 case); E-103 68 %, E-104 51 %. The model's exchangers are smaller than the datasheets.	Accept for training (the model reproduces the design point); note that the datasheet over-surface is spent at 6 % CO2.

Blocked-outlet basis: Blocked CO2 outlet: PIC-301 in MAN with PV-301 closed; every trip disabled, from the design state (SRS of SIF tab, MEA-IN-SRS-001). The model clamps the drum pressure at 400 kPa a, so the true peak is at least this.

The tab "Mechanical" of the OTS recomputes this equipment list in the page and checks it against the model, the PSV and trip settings, the pump shut-off pressures and the service (✓/✗).